

Multinomial Logit Model

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This assignment explores two methods for estimating the MNL model: (1) via Maximum Likelihood, and (2) via a Bayesian approach using a Metropolis-Hastings MCMC algorithm.

1. Likelihood for the Multi-nomial Logit (MNL) Model

Suppose we have $i = 1, \dots, n$ consumers who each select exactly one product j from a set of J products. The outcome variable is the identity of the product chosen $y_i \in \{1, \dots, J\}$ or equivalently a vector of $J - 1$ zeros and 1 one, where the 1 indicates the selected product. For example, if the third product was chosen out of 3 products, then either $y = 3$ or $y = (0, 0, 1)$ depending on how we want to represent it. Suppose also that we have a vector of data on each product x_j (eg, brand, price, etc.).

We model the consumer's decision as the selection of the product that provides the most utility, and we'll specify the utility function as a linear function of the product characteristics:

$$U_{ij} = x'_j \beta + \epsilon_{ij}$$

where ϵ_{ij} is an i.i.d. extreme value error term.

The choice of the i.i.d. extreme value error term leads to a closed-form expression for the probability that consumer i chooses product j :

$$\mathbb{P}_i(j) = \frac{e^{x'_j \beta}}{\sum_{k=1}^J e^{x'_k \beta}}$$

For example, if there are 3 products, the probability that consumer i chooses product 3 is:

$$\mathbb{P}_i(3) = \frac{e^{x'_3 \beta}}{e^{x'_1 \beta} + e^{x'_2 \beta} + e^{x'_3 \beta}}$$

A clever way to write the individual likelihood function for consumer i is the product of the J probabilities, each raised to the power of an indicator variable (δ_{ij}) that indicates the chosen product:

$$L_i(\beta) = \prod_{j=1}^J \mathbb{P}_i(j)^{\delta_{ij}} = \mathbb{P}_i(1)^{\delta_{i1}} \times \dots \times \mathbb{P}_i(J)^{\delta_{iJ}}$$

Notice that if the consumer selected product $j = 3$, then $\delta_{i3} = 1$ while $\delta_{i1} = \delta_{i2} = 0$ and the likelihood is:

$$L_i(\beta) = \mathbb{P}_i(1)^0 \times \mathbb{P}_i(2)^0 \times \mathbb{P}_i(3)^1 = \mathbb{P}_i(3) = \frac{e^{x'_3\beta}}{\sum_{k=1}^3 e^{x'_k\beta}}$$

The joint likelihood (across all consumers) is the product of the n individual likelihoods:

$$L_n(\beta) = \prod_{i=1}^n L_i(\beta) = \prod_{i=1}^n \prod_{j=1}^J \mathbb{P}_i(j)^{\delta_{ij}}$$

And the joint log-likelihood function is:

$$\ell_n(\beta) = \sum_{i=1}^n \sum_{j=1}^J \delta_{ij} \log(\mathbb{P}_i(j))$$

2. Simulate Conjoint Data

We will simulate data from a conjoint experiment about video content streaming services. We elect to simulate 100 respondents, each completing 10 choice tasks, where they choose from three alternatives per task. For simplicity, there is not a “no choice” option; each simulated respondent must select one of the 3 alternatives.

Each alternative is a hypothetical streaming offer consistent of three attributes: (1) brand is either Netflix, Amazon Prime, or Hulu; (2) ads can either be part of the experience, or it can be ad-free, and (3) price per month ranges from \$4 to \$32 in increments of \$4.

The part-worths (ie, preference weights or beta parameters) for the attribute levels will be 1.0 for Netflix, 0.5 for Amazon Prime (with 0 for Hulu as the reference brand); -0.8 for included advertisements (0 for ad-free); and $-0.1 \times \text{price}$ so that utility to consumer i for hypothetical streaming service j is

$$u_{ij} = (1 \times \text{Netflix}_j) + (0.5 \times \text{Prime}_j) + (-0.8 \times \text{Ads}_j) - 0.1 \times \text{Price}_j + \varepsilon_{ij}$$

where the variables are binary indicators and ε is Type 1 Extreme Value (ie, Gumble) distributed.

The following code provides the simulation of the conjoint data.

R code on ChatGPT

3. Preparing the Data for Estimation

The “hard part” of the MNL likelihood function is organizing the data, as we need to keep track of 3 dimensions (consumer i , covariate k , and product j) instead of the typical 2 dimensions for cross-sectional regression models (consumer i and covariate k). The fact that each task for each respondent has the same number of alternatives (3) helps. In addition, we need to convert the categorical variables for brand and ads into binary variables.

todo: reshape and prep the data

4. Estimation via Maximum Likelihood

todo: Code up the log-likelihood function.

todo: Use `optim()` in R or `scipy.optimize()` in Python to find the MLEs for the 4 parameters (β_{netflix} , β_{prime} , β_{ads} , β_{price}), as well as their standard errors (from the Hessian). For each parameter construct a 95% confidence interval.

5. Estimation via Bayesian Methods

todo: code up a metropolis-hasting MCMC sampler of the posterior distribution. Take 11,000 steps and throw away the first 1,000, retaining the subsequent 10,000.

hint: Use $N(0,5)$ priors for the betas on the binary variables, and a $N(0,1)$ prior for the price beta.

__hint: instead of calculating $\text{post}=\text{lik}*\text{prior}$, you can work in the log-space and calculate $\log\text{-post} = \log\text{-lik} + \log\text{-prior}$ (this should enable you to re-use your log-likelihood function from the MLE section just above)__

hint: King Markov (in the video) use a candidate distribution of a coin flip to decide whether to move left or right among his islands. Unlike King Markov, we have 4 dimensions (because we have 4 betas) and our dimensions are continuous. So, use a multivariate normal distribution to propose the next location for the algorithm to move to. I recommend a $MNV(\mu, \Sigma)$ where $\mu=c(0,0,0,0)$ and Σ has diagonal values $c(0.05, 0.05, 0.05, 0.005)$ and zeros on the off-diagonal. Since this MVN has no covariances, you can sample each dimension independently

(so 4 univariate normals instead of 1 multivariate normal), where the first 3 univariate normals are $N(0, 0.05)$ and the last one is $N(0, 0.005)$.

todo: for at least one of the 4 parameters, show the trace plot of the algorithm, as well as the histogram of the posterior distribution.

todo: report the 4 posterior means, standard deviations, and 95% credible intervals and compare them to your results from the Maximum Likelihood approach.

6. Discussion

todo: Suppose you did not simulate the data. What do you observe about the parameter estimates? What does $\beta_{\text{Netflix}} > \beta_{\text{Prime}}$ mean? Does it make sense that β_{price} is negative?

todo: At a high level, discuss what change you would need to make in order to simulate data from — and estimate the parameters of — a multi-level (aka random-parameter or hierarchical) model. This is the model we use to analyze “real world” conjoint data.